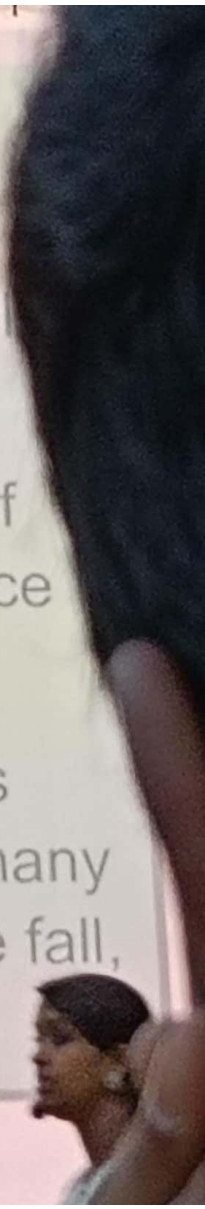
A scenario...

You design websites for local businesses. You charge $200 per website, and currently se 12 websites per month.

Your costs are rising (including the opp. cost of your time), so you're thinking of raising the price to $250.

The law of demand says that you won't sell as many websites if you raise your price. How many fewer websites? How much will your revenue fall, or might it increase?



Elasticity

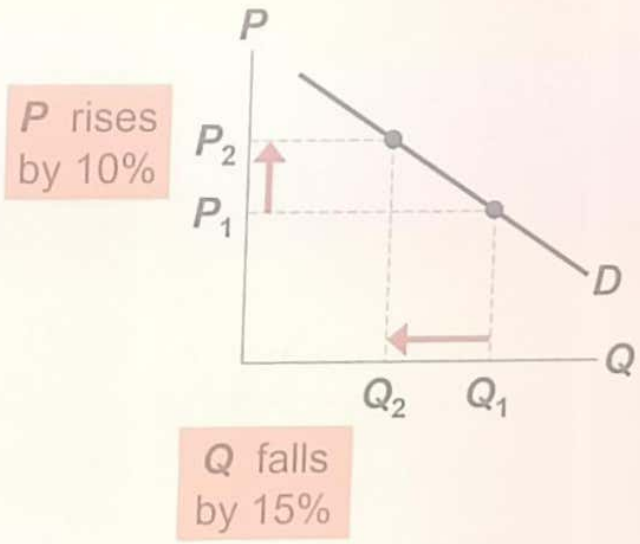
- Basic idea: Elasticity measures how much one variable responds to changes in another variable.
  - One type of elasticity measures how much demand for your websites will fall if you raise your price.

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

Example:

Price elasticity of demand equals

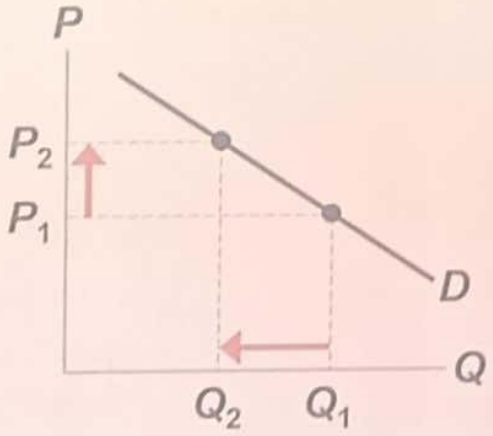


Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

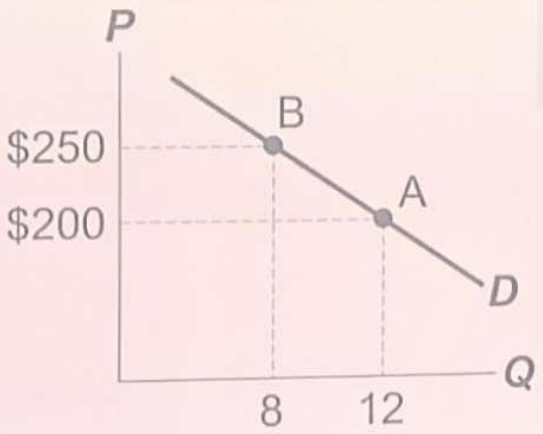
Along a ***D*** curve, ***P*** and ***Q*** move in opposite directions, which would make price elasticity negative.

We will drop the minus sign and report all price elasticities as positive numbers.



Calculating Percentage Changes

Demand for your websites

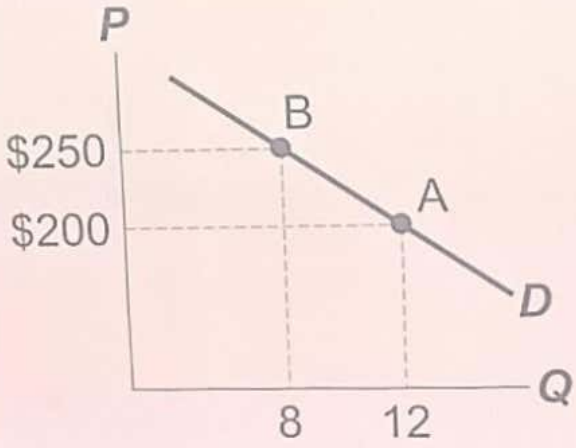


Standard method of computing the percentage (%) change:

$$\frac{\text{end value} - \text{start value}}{\text{start value}} \times 100\%$$

Calculating Percentage Changes

Demand for your websites



Problem:

The standard method gives different answers depending on where you start.

From A to B,
P rises 25%, ***Q*** falls 33%,
elasticity = 33/25 = 1.33

From B to A,
P falls 20%, ***Q*** rises 50%,
elasticity = 50/20 = 2.50

Calculating Percentage Changes

- So, we instead use the **midpoint method**:

$$\frac{\text{end value} - \text{start value}}{\text{midpoint}} \times 100\%$$

- The midpoint is the number halfway between the start & end values, also the average of those values.

Calculating Percentage Changes

- Using the midpoint method, the % change in ***P*** equals

$$\frac{\$250 - \$200}{\$225} \times 100\% = 22.2\%$$

- The % change in **Q** equals

$$\frac{12 - 8}{10} \times 100\% = 40.0\%$$

- The price elasticity of demand equals

$$40/22.2 = 1.8$$

ACTIVE LEARNING 1:
Calculate an elasticity

Use the following information to calculate the price elasticity of demand for hotel rooms:

if P = \$70, Q^d = 5000

if P = \$90, Q^d = 3000

What determines price elasticity?

To learn the determinants of price elasticity, we look at a series of examples.
Each compares two common goods.

In each example:

- Suppose the prices of both goods rise by 20%.
- The good for which Q^d falls the most (in percent) has the highest price elasticity of demand. Which good is it? Why?
- What lesson does the example teach us about determinants of the price elasticity of dema

EXAMPLE 1:
Potato chips vs. Sunscreen

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - Potato Chips has lots of close substitutes (*e.g.*, other chips- Kurkure, Biscuits, etc), so buyers can easily switch if the price rises.
  - Sunscreen has no close substitutes, so consumers would probably not buy much less if its price rises.
- Lesson: ***Price elasticity is higher when close substitutes are available.***

EXAMPLE 2:
"Tata Nexon" vs. "Cars"

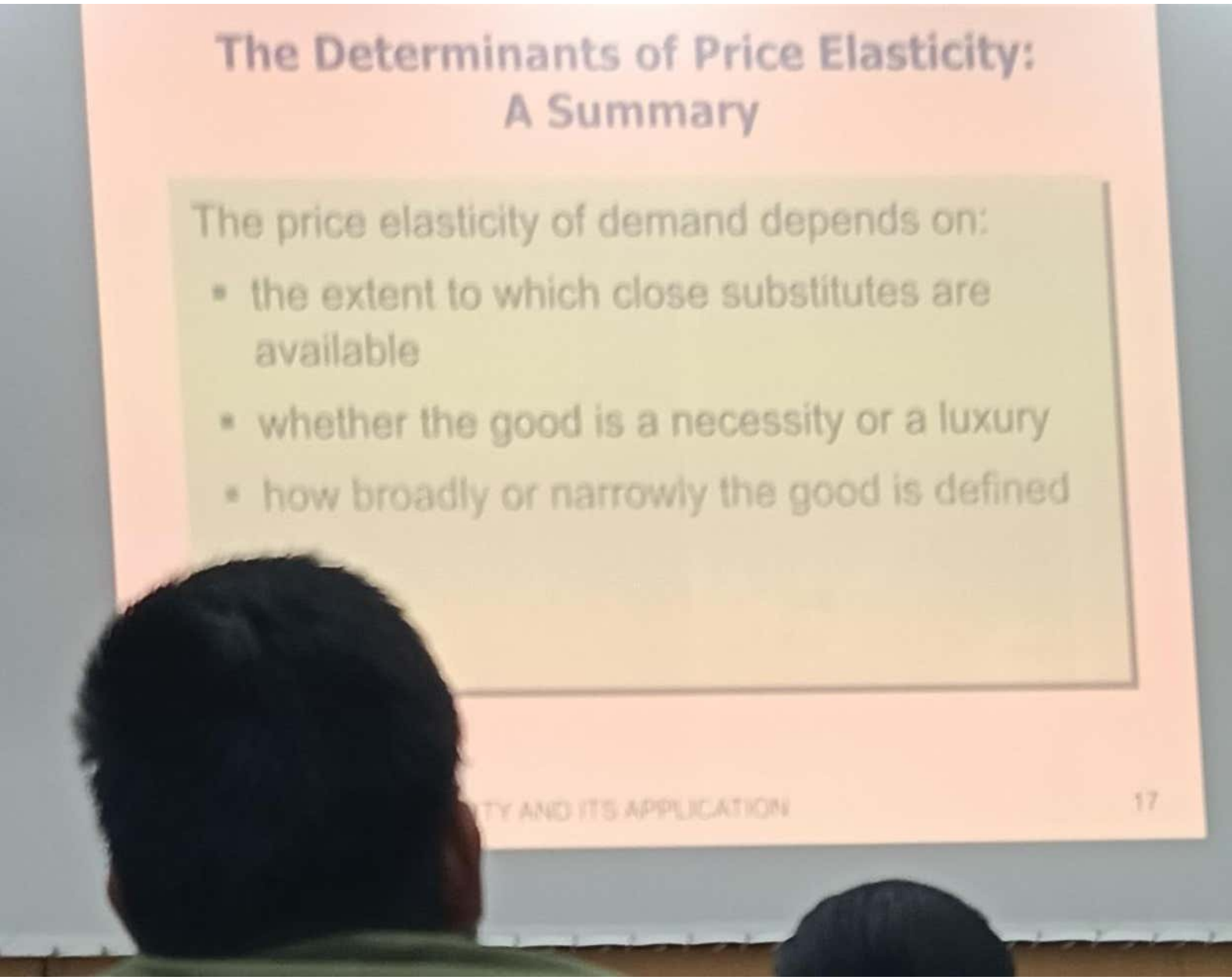
- The prices of both goods rise by 20%. For which good does Q^d drop the most? Why?
  - For a narrowly defined good such as a car brand/model, there are many substitutes.
  - There are fewer substitutes available for broadly defined goods such as cars.
  - Lesson: *Price elasticity is higher for narrowly defined goods than broadly defined ones.*

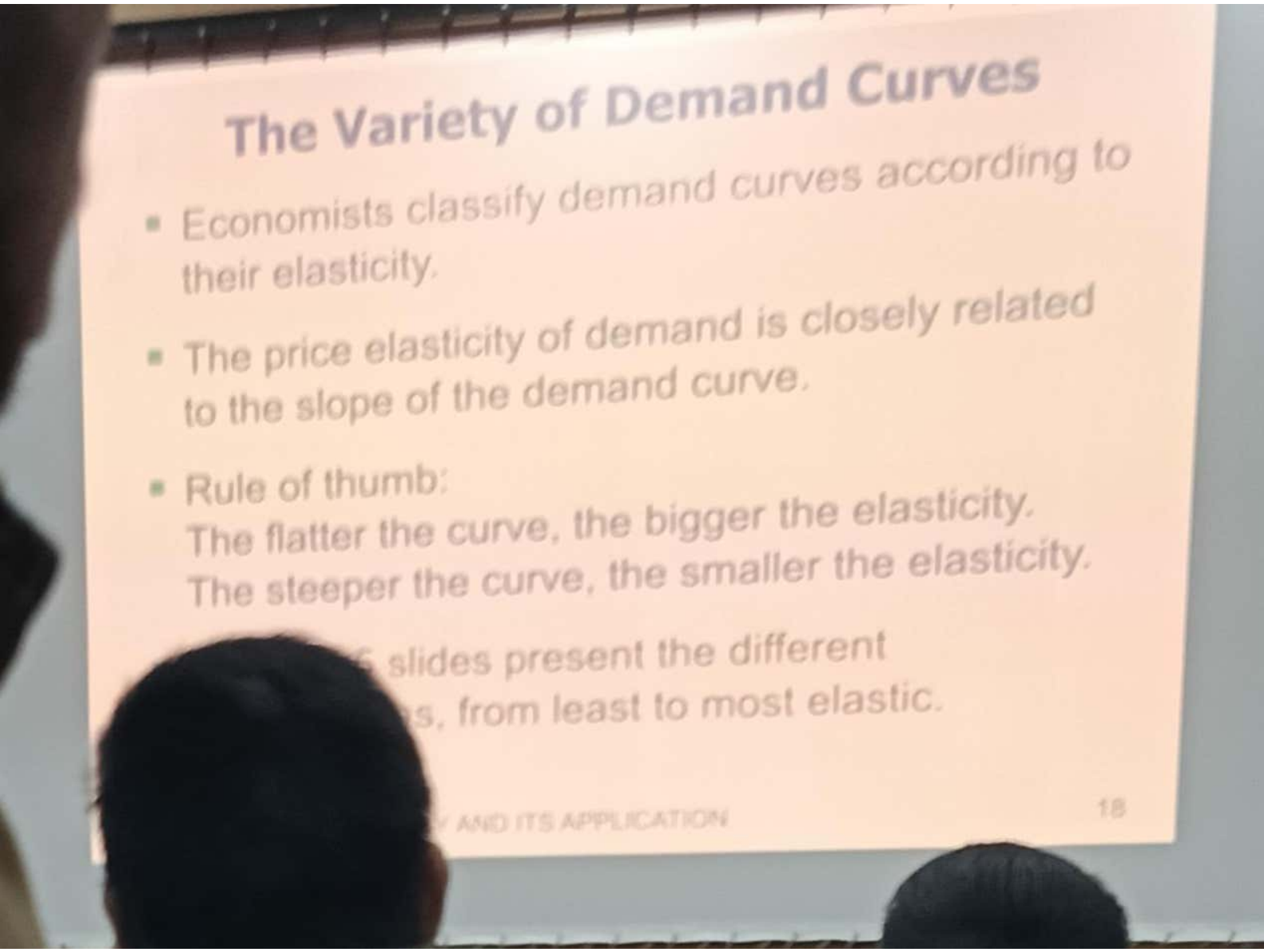
EXAMPLE 3:
Insulin vs. Cruises

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - To millions of diabetics, insulin is a necessity. A rise in its price would cause little or no decrease in demand.
  - A cruise is a luxury. If the price rises, some people will forego it.
- Lesson: ***Price elasticity is higher for luxuries than for necessities.***

EXAMPLE 4:
Gasoline in the Short Run vs. Gasoline in the Long Run

- The price of gasoline rises 20%. Does Q^d drop more in the short run or the long run? Why?
  - There's not much people can do in the short run, other than ride the bus or carpool.
  - In the long run, people can buy smaller cars or live closer to where they work.
- Lesson: ***Price elasticity is higher in the long run than the short run.***


The Determinants of Price Elasticity:
A Summary
The price elasticity of demand depends on:
• the extent to which close substitutes are available
• whether the good is a necessity or a luxury
• how broadly or narrowly the good is defined
TY AND ITS APPLICATION
17


The Variety of Demand Curves
Economists classify demand curves according to their elasticity.
The price elasticity of demand is closely related to the slope of the demand curve.
Rule of thumb:
The flatter the curve, the bigger the elasticity.
The steeper the curve, the smaller the elasticity.
slides present the different
s, from least to most elastic.
AND ITS APPLICATION
18

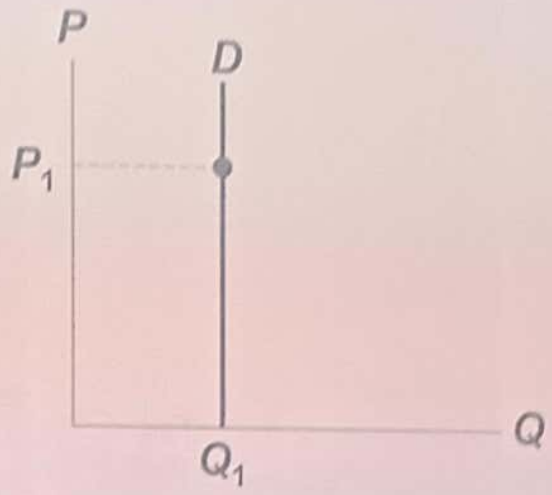
"Perfectly inelastic demand" (one extreme case)

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } Q}{\%\ \text{change in } P} = \frac{\quad}{\quad}$$

D curve:
vertical

Consumers' price sensitivity:
0

Elasticity:



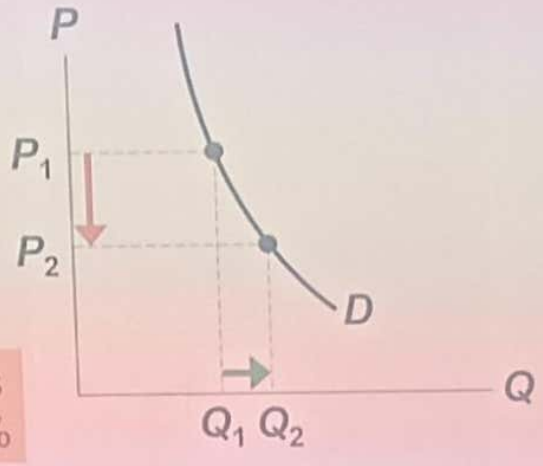
"Inelastic demand"

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{< 10\%}{10\%} < 1$$

D curve:
relatively steep

Consumers' price sensitivity:
relatively low

Elasticity:
< 1



P falls by 10%

Q rises less than 10%

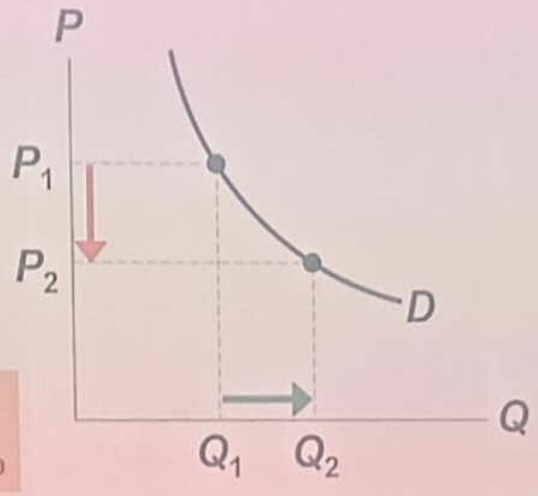
"Unit elastic demand"

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{10\%}{10\%} = 1$$

D curve:
intermediate slope

Consumers' price sensitivity:
intermediate

Elasticity:
1



P falls by 10%

Q rises by 10%

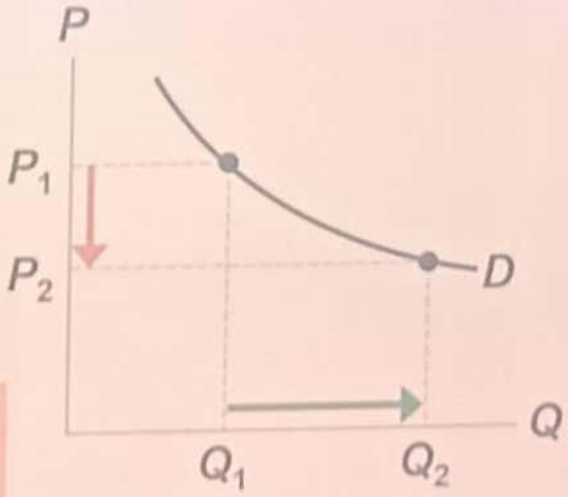
"Elastic demand"

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{> 10\%}{10\%} > 1$$

D curve:
relatively flat

Consumers' price sensitivity:
relatively high

Elasticity:
> 1



P falls by 10%

Q rises more than 10%

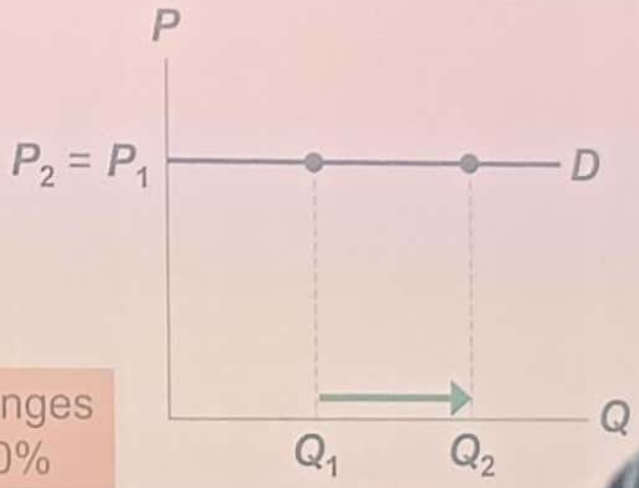
"Perfectly elastic demand" (the other extreme)

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } Q}{\%\ \text{change in } P} = \frac{\text{any } \%}{0\%}$$

D curve:
horizontal

Consumers' price sensitivity:
extreme

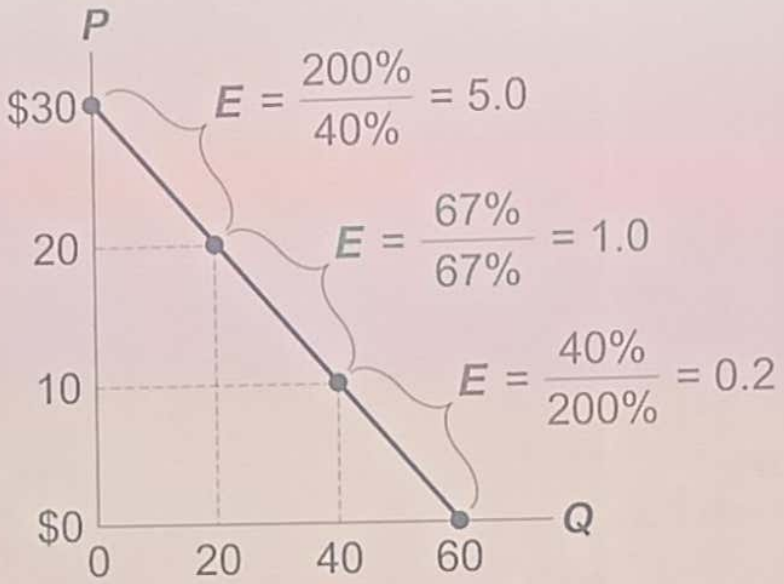
Elasticity:



P changes by 0%

Q changes by any %

Elasticity of a Linear Demand Curve



The slope of a linear demand curve is constant, but its elasticity is not.

Price Elasticity and Total Revenue

- Continuing our scenario, if you raise your price from $200 to $250, would your revenue rise or fall?

$$\text{Revenue} = P \times Q$$

- A price increase has two effects on revenue:
  - Higher *P* means more revenue on each unit you sell.
  - But you sell fewer units (lower *Q*), due to Law of Demand.

Price Elasticity and Total Revenue

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q}{\text{Percentage change in } P}$$

$$\text{Revenue} = P \times Q$$

- If demand is elastic, then

  price elast. of demand > 1

  % change in Q > % change in P

- The fall in revenue from lower Q is greater than the increase in revenue from higher P, so revenue falls.

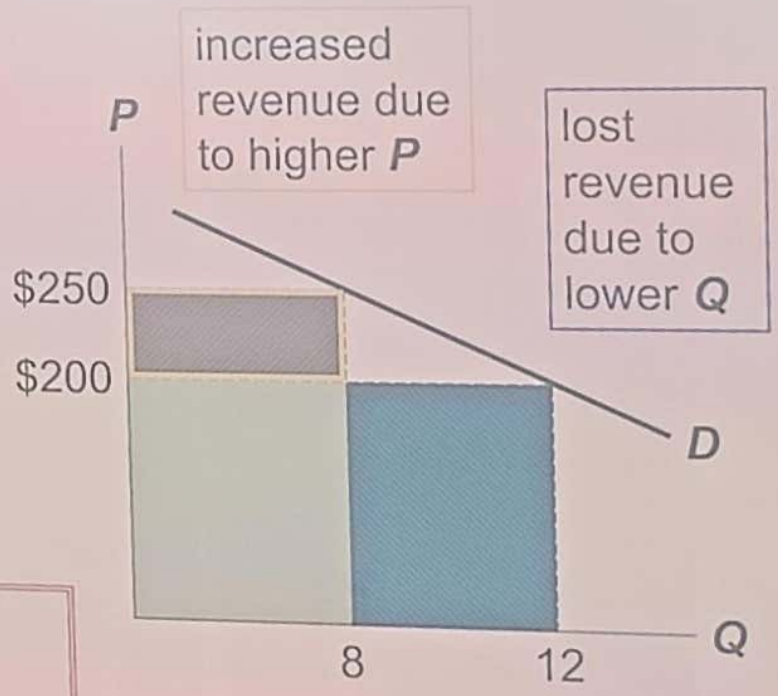
Price Elasticity and Total Revenue

Elastic demand
(elasticity = 1.8)

If P = \$200,
Q = 12 and
revenue = \$2400.

If P = \$250,
Q = 8 and
revenue = \$2000.

When D is elastic,
a price increase
causes revenue to fall.



Price Elasticity and Total Revenue

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q}{\text{Percentage change in } P}$$

Revenue = $P \times Q$

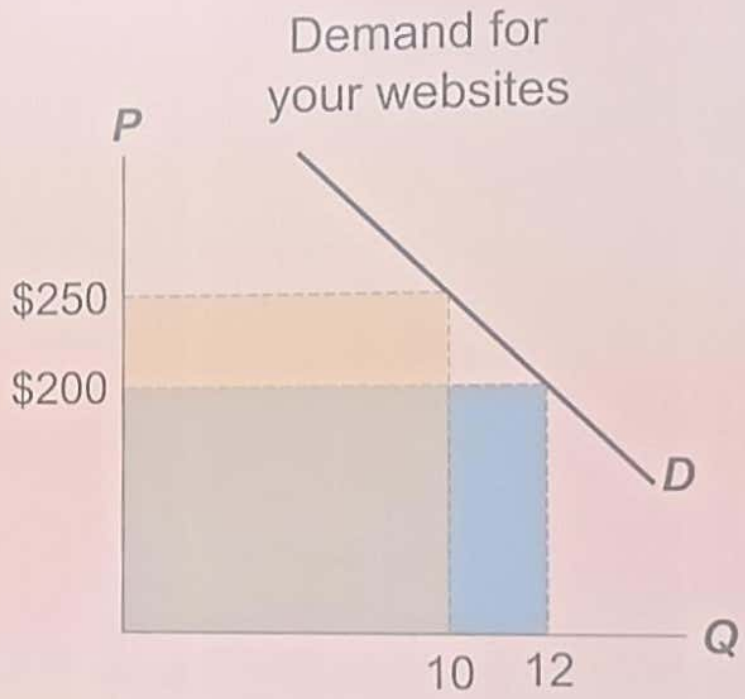
- If demand is inelastic, then
  price elast. of demand < 1
  % change in Q $<$ % change in P
- The fall in revenue from lower Q is smaller than the increase in revenue from higher P, so revenue rises.
- In our example, suppose that Q only falls to 10 (instead of 8) when you raise your price to \$250.

Price Elasticity and Total Revenue

Now, demand is inelastic:
elasticity = 0.82

If P = \$200, Q = 12 and revenue = \$2400.

If P = \$250, Q = 10 and revenue = \$2500.



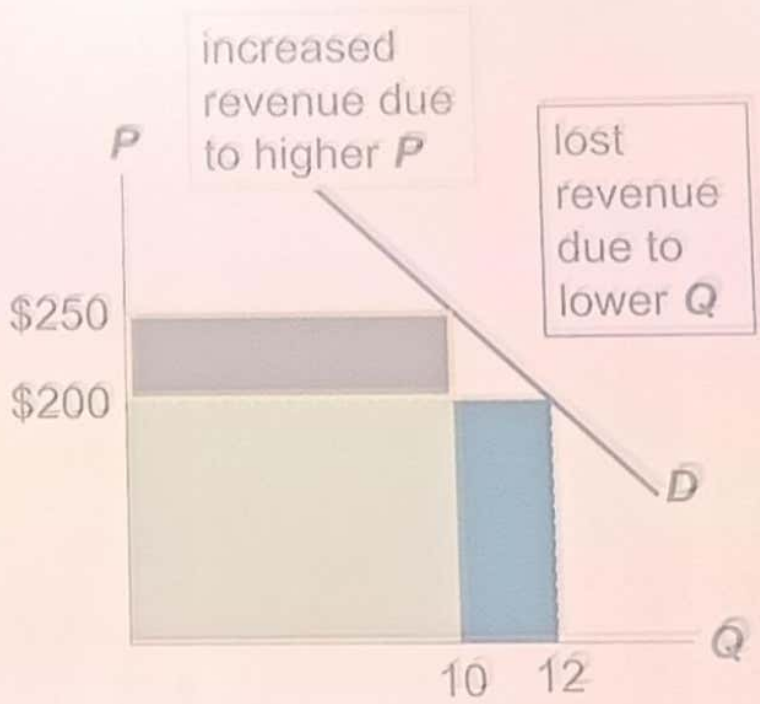
Price Elasticity and Total Revenue

Now, demand is inelastic:

elasticity = 0.82

If P = \$200, Q = 12 and revenue = \$2400.

If P = \$250, Q = 10 and revenue = \$2500.



ACTIVE LEARNING 2:

Elasticity and expenditure/revenue

A. Pharmacies raise the price of insulin by 10%.
Does total expenditure on insulin rise or fall?

B. As a result of a fare war, the price of a luxury cruise falls 20%.
Does luxury cruise companies' total revenue rise or fall?

ACTIVE LEARNING 2:
Answers

A. Pharmacies raise the price of insulin by 10%. Does total expenditure on insulin rise or fall?

Expenditure = ***P*** x ***Q***

Since demand is inelastic, ***Q*** will fall less than 10%, so expenditure rises.

ACTIVE LEARNING 2:
Answers

B. As a result of a fare war, the price of a luxury cruise falls 20%.
Does luxury cruise companies' total revenue rise or fall?

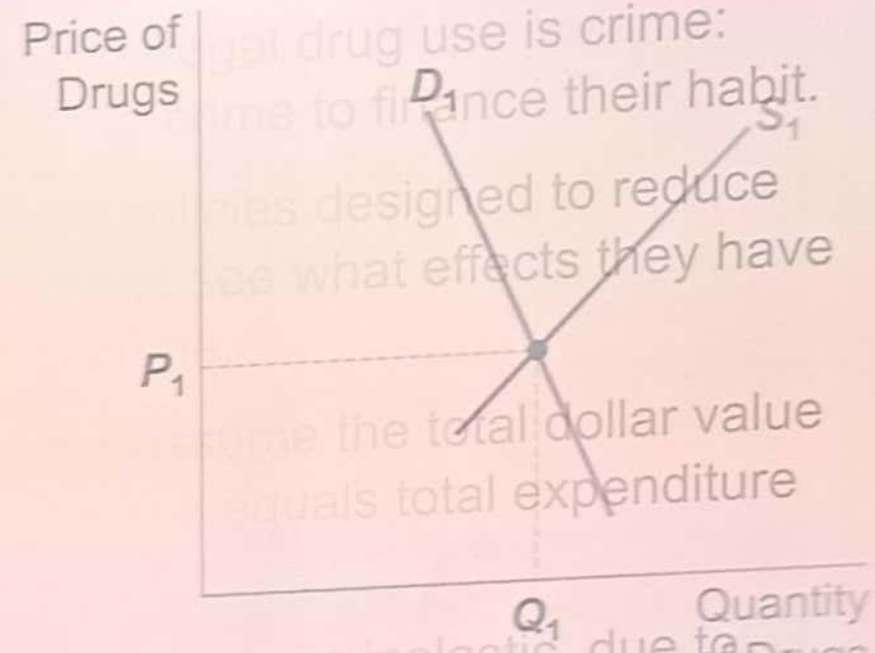
Revenue = $P \times Q$

The fall in P reduces revenue,
but Q increases, which increases revenue.
Which effect is bigger?

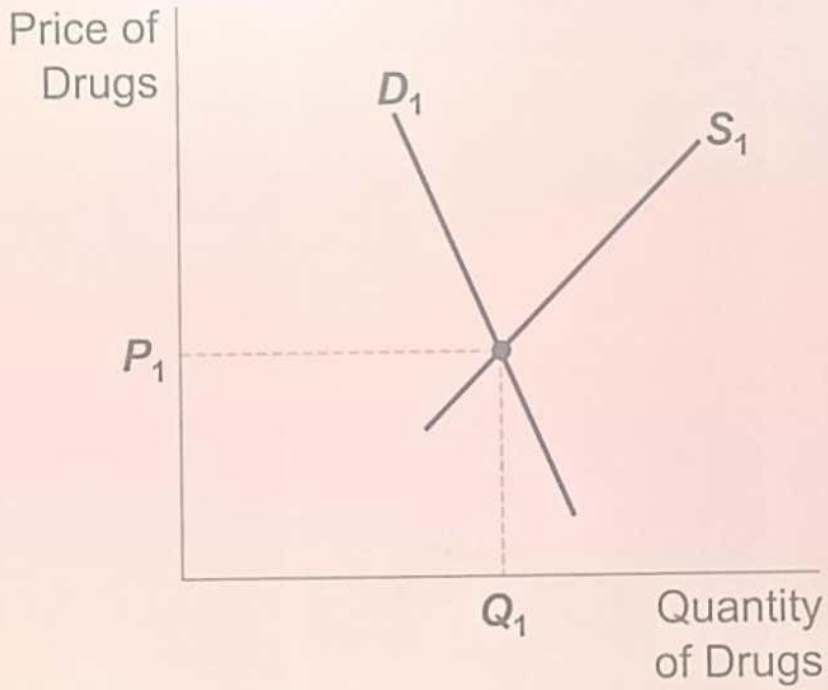
Since demand is elastic, Q will increase more than 20%, so revenue rises.

APPLICATION: Does Drug Interdiction Increase or Decrease Drug-Related Crime?

Policy 1: Interdiction



Policy 1: Interdiction

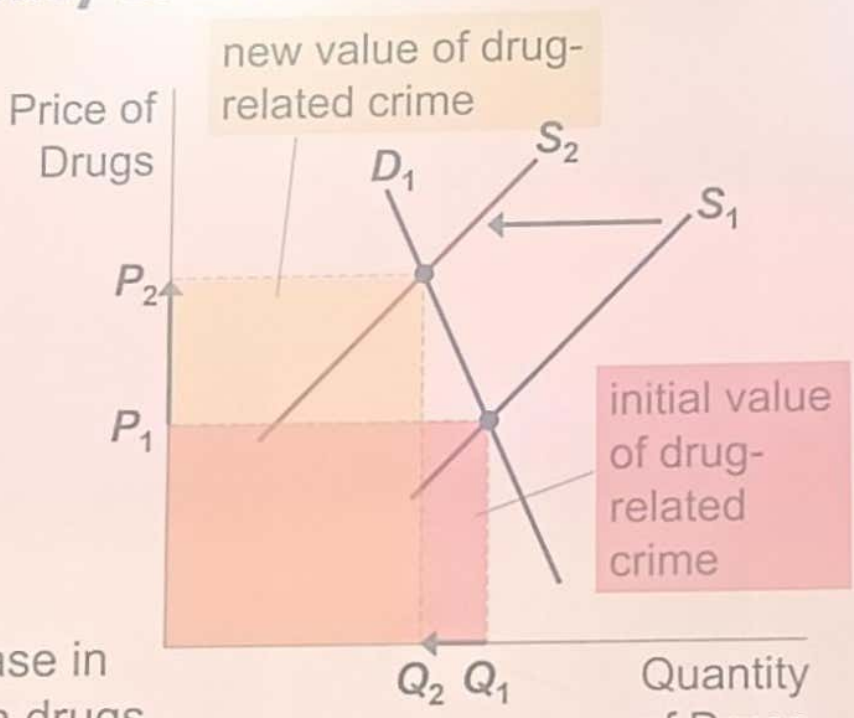


Policy 1: Interdiction

Interdiction reduces the supply of drugs.

Since demand for drugs is inelastic, *P* rises proportionally more than *Q* falls.

Result: an increase in total spending on drugs, and in drug-related crime

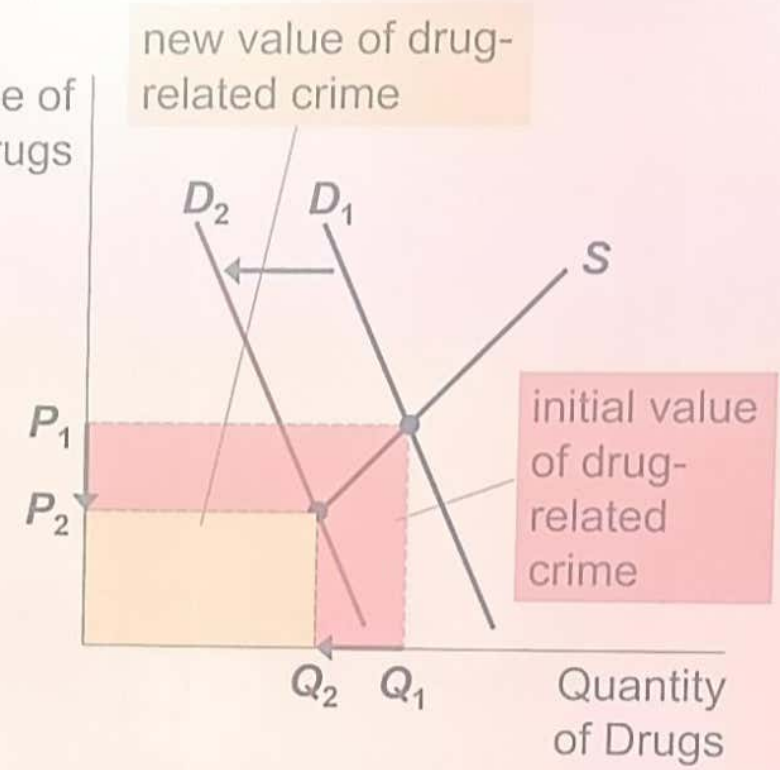


Policy 2: Education

Education reduces the demand for drugs.

P and *Q* fall.

Result:
A decrease in total spending on drugs, and in drug-related crime.



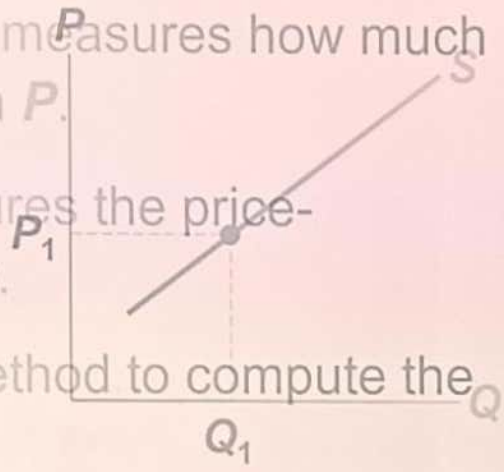
Price Elasticity of Supply

- **Price elasticity of supply** measures how much Q^S responds to a change in P.

Price Elasticity of Supply

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^s}{\text{Percentage change in } P}$$

Example:



Price Elasticity of Supply

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^s}{\text{Percentage change in } P}$$

- **Price elasticity of supply** measures how much Q^s responds to a change in P.
- Loosely speaking, it measures the price-sensitivity of sellers' supply.
- Again, use the midpoint method to compute the percentage changes.

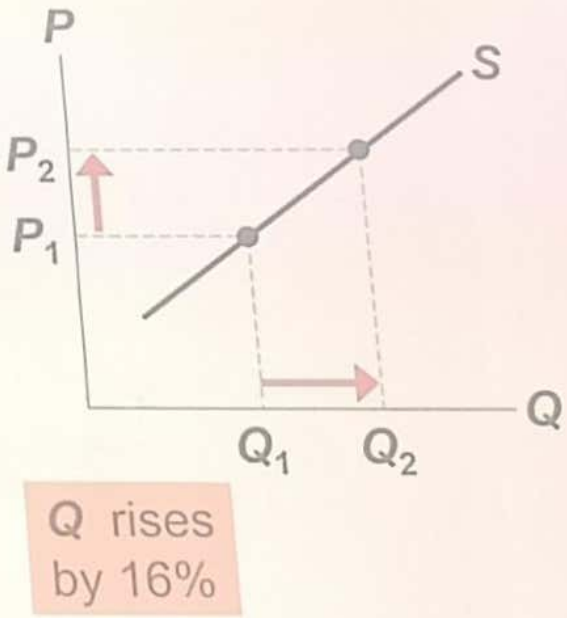
Price Elasticity of Supply

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^S}{\text{Percentage change in } P}$$

Example:

Price elasticity of supply equals

$$\frac{16\%}{8\%} = 2.0$$



The Variety of Supply Curves

- Economists classify supply curves according to their elasticity.
- The slope of the supply curve is closely related to price elasticity of supply.
- Rule of thumb:
  The flatter the curve, the bigger the elasticity.
  The steeper the curve, the smaller the elasticity.
- The next 5 slides present the different classifications, from least to most elastic.

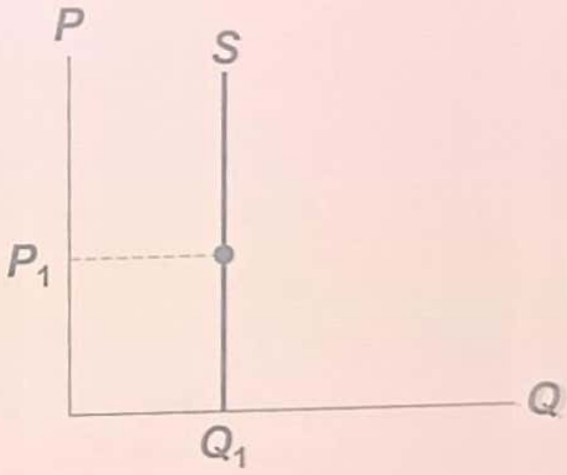
"Perfectly inelastic" (one extreme)

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \text{_____}$$

S curve:
vertical

Sellers' price sensitivity:
0

Elasticity:



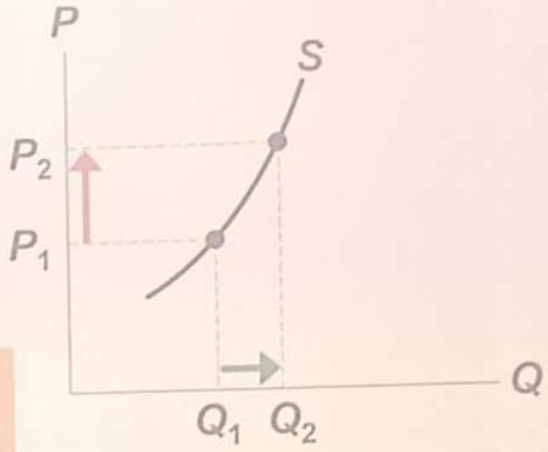
"Inelastic"

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{< 10\%}{10\%}$$

S curve:
relatively steep

Sellers' price sensitivity:
relatively low

Elasticity:



P rises by 10%

Q rises less than 10%

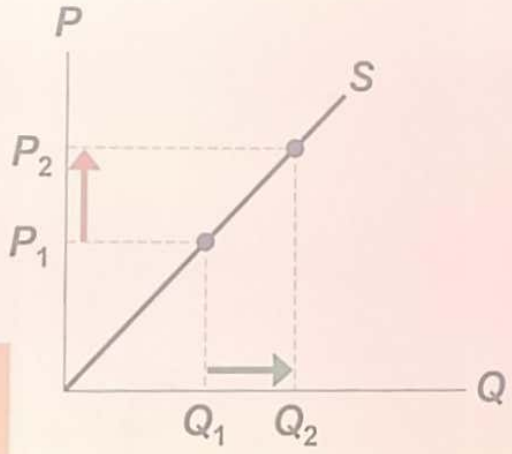
"Unit elastic"

$$\text{Price elasticity of supply} = \frac{\%\ \text{change in } Q}{\%\ \text{change in } P} = \frac{10\%}{10\%} = 1$$

S curve:
intermediate slope

Sellers' price sensitivity:
intermediate

Elasticity:
= 1



P rises by 10%

Q rises by 10%

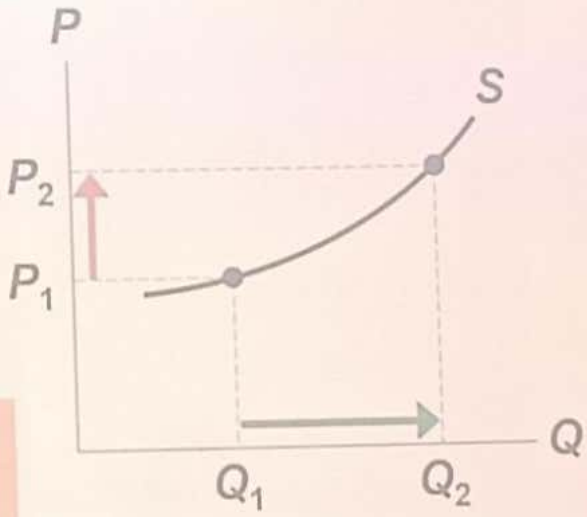
"Elastic"

$$\text{Price elasticity of supply} = \frac{\%\ \text{change in } Q}{\%\ \text{change in } P} = \frac{> 10\%}{10\%}$$

S curve:
relatively flat

Sellers' price sensitivity:
relatively high

Elasticity:



P rises by 10%

Q rises more than 10%

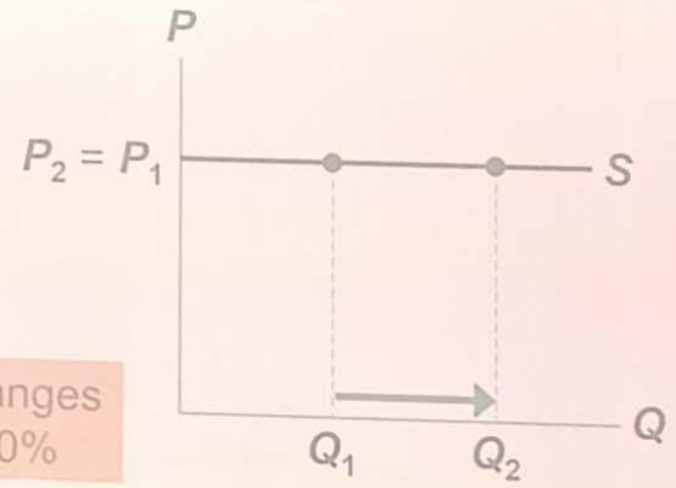
"Perfectly elastic" (the other extreme)

$$\text{Price elasticity of supply} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{\text{any }\%}{0\%} = \text{infinity}$$

S curve:
horizontal

Sellers' price sensitivity:
extreme

Elasticity:
infinity



P changes by 0%

Q changes by any %

The Determinants of Supply Elasticity

- The more easily sellers can change the quantity they produce, the greater the price elasticity of supply.
- Example: Supply of beachfront property is harder to vary and thus less elastic than supply of new cars.
- For many goods, price elasticity of supply is greater in the long run than in the short run, because firms can build new factories, or new firms may be able to enter the market.

ACTIVE LEARNING 3:

Elasticity and changes in equilibrium

- The supply of beachfront property is inelastic.
  The supply of new cars is elastic.